

ASM-ESD 101.1.0

Advanced SPICE Model for ESD Diodes

Technical Manual

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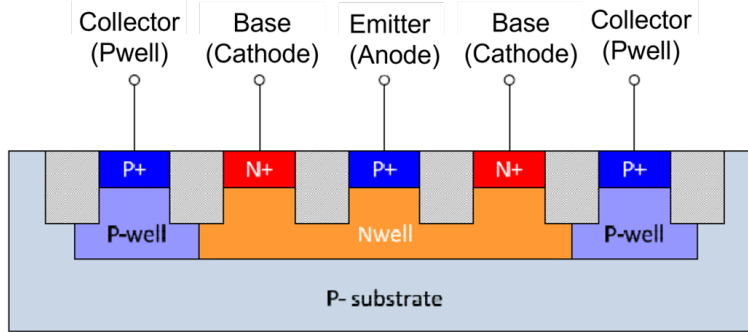


Figure 1: Typical ESD diode cross-section showing anode, cathode, and P+ well terminals of the device.

1 ASM-ESD Overview

Electro-static discharge (ESD) diodes are the most popular component for protection of core electronics against any potentially harmful ESD events. Accurate circuit simulation of the ESD diode behavior is thus critical for designers to be able to analyze its effect on the overall circuit performance. Accurate simulations are also needed to perform any co-optimization of ESD protection devices and the core circuits. The accuracy of circuit simulations is primarily dependent on the accuracy and capabilities of the compact/SPICE model.

A compact model for ESD behavior needs to include a plethora of physical effects. These are necessary for accuracy under various ESD conditions. Some of these physical effects include modeling of parasitic BJT in diodes, advanced approach for modeling the self-heating effect, modeling of transient or overshoot effects, parasitic BJT roll-off effect and an accurate modeling of post reverse breakdown I-V behavior. These effects should also be captured across multiple ambient temperatures. Advance SPICE model for ESD diodes (ASM-ESD) is a single comprehensive model with ability to capture all these important effects for ESD simulations is not available. This technical manual describes ASM-ESD model in detail.

A cross-section view of ESD diodes manufactured in a typical manufacturing process is shown in Fig. 1. In addition to the anode and the cathode terminals, a third terminal P+ well can be seen in Fig. 1. ESD diode is thus a three terminal device where the P+ well terminal is equivalent to a collector terminal of a bipolar junction transistor (BJT). The current flowing to collector/P+ well terminal can be significant under ESD conditions. This parasitic BJT effect is modeled in ASM-ESD model with

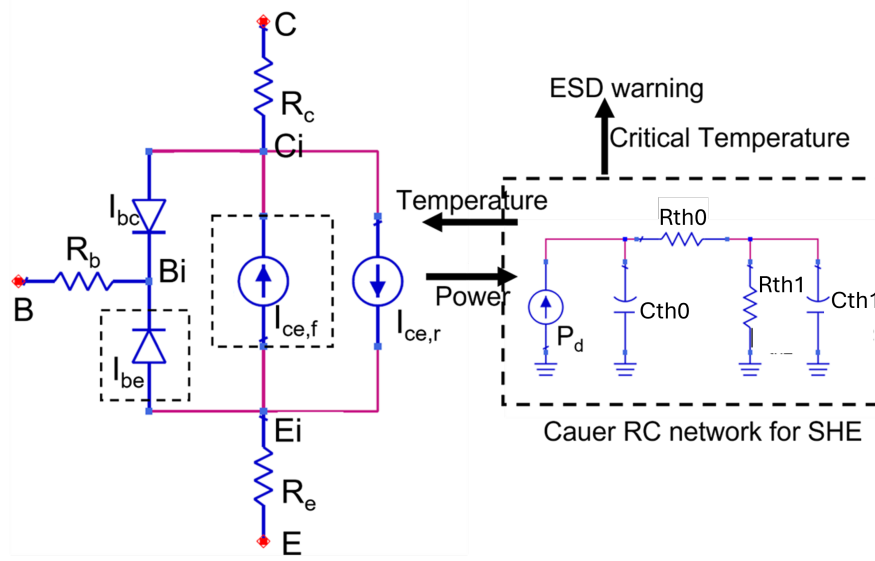


Figure 2: ASM-ESD diode model topology. C represents substrate connection in a diode, B and E represent the anode or cathode of the diode.

specific enhancements to the model formulations of the standard Gummel-Poon (SGP) BJT model.

ASM-ESD model consist of core formulations which model the forward- and the reverse I-V diode behavior under normal operation. The core model formulations are augmented with the ability to model a plethora of ESD-specific physical effects.

ASM-ESD model topology is shown in Fig. 2. The basic topology resembles that of the Gummel-Poon model [11]. The dependent current sources model the parasitic BJT in the diode. Several ESD-specific enhancements to the original topology and the formulations have been made. The enhancements to the model topology include a bias-dependent base resistance model for ESD events, a multi-pole thermal network for self-heating effect and failure temperature warning. The added new formulations to account for ESD-specific effects include enhancements related to the transient overshoot effects, the post reverse breakdown resistance, and the delayed transient response model for the parasitic BJT gain. All the formulations of ASM-ESD model have been implemented in Verilog-AMS [12] programming language. ASM-ESD model can be simulated in all the major circuit simulators. Different parts of the model are discussed in the following sub-sections.

DTYPE = 1 $\rightarrow$ NPN
DTYPE = -1 $\rightarrow$ PNP

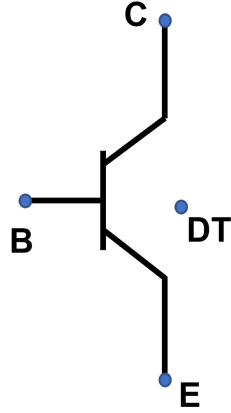


Figure 3: ASM-ESD has four terminals. C represents substrate connection in a diode, B and E represent the anode or cathode of the diode. Both NPN and PNP configurations can be modeled by setting model parameter DTYPE.

1.1 ASM-ESD model terminals

ASM-ESD models the ESD diode in two configurations: 1. Four terminal and 2. Two terminal. In a four terminal device as shown in Fig. 3. Collector (C), Base (B), and Emitter (E) are electrical terminals. Terminal DT is the thermal node which can be left floating or can be connected to an extrinsic thermal network as required by the user. For ESD diodes the terminals B and E represent the anode(cathode) of the diode for a NPN(PNP) device. C represents substrate connection. In a two terminal configuration B and E are the nodes used by the model.

2 ASM-ESD 101.1.0 Model Equations

Physical constants, voltage equations, temperature dependent equations, charge and basic current equations are presented in this section. Note that the model parameters are indicated in capital letters in the equations below and in the parameter tables in the manual for better readability. The parameters are implemented in small letters in the Verilog-A code.

2.1 Physical Constants

Physical quantities in ASM-ESD are in S.I. units unless specified otherwise.

$$q = 1.6021918 \times 10^{-19}$$

$$E_g = 1.1150877$$

$$K_B = 1.3806226 \times 10^{-23}$$

$$\frac{K_B}{q} = 8.617086 \times 10^{-5}$$

$$egta = 7.02000 \times 10^{-4}$$

$$egtb = 1.10800 \times 10^3$$

$$Ref_t = 300.1500$$

2.2 Extrinsic and intrinsic voltages

Intrinsic terminal voltages is calculated based on the applied external voltage at terminals C, B, and E and the amount of current in each terminal according to the following equations.

$$V_{bi} = V_b - R_b \cdot I_b \tag{2.2.1}$$

$$V_{ci} = V_c - R_c \cdot I_c \quad (2.2.2)$$

$$V_{ei} = V_e - R_e \cdot I_e \quad (2.2.3)$$

where R_b , R_c and R_e are base-, collector, and emitter resistances respectively.

Intrinsic junction voltages V_{bei} and V_{bci} are calculated from the above intrinsic voltages according to the following equations.

$$V_{bei} = V_{bi} - V_{ei} \quad (2.2.4)$$

$$V_{bci} = V_{bi} - V_{ci} \quad (2.2.5)$$

2.3 Temperature dependent calculations

The device temperature T_{dev} is calculated as,

$$T_{dev} = T_{amb} + Temp(dt) + DTEMP \quad (2.3.1)$$

$$V_t = KBoQ \cdot T_{dev} \quad (2.3.2)$$

$$rT = \frac{T_{dev}}{TNOM} \quad (2.3.3)$$

$$\ln rT = \ln(rT) \quad (2.3.4)$$

Temperature dependence of BJT forward (BF) and reverse gain (BR) is given by,

$$BF_t = BF \cdot \exp(XTB \cdot \ln rT) \quad (2.3.5)$$

$$BR_t = BR \cdot \exp(XTB \cdot \ln rT) \quad (2.3.6)$$

$$argt = XTI \cdot \ln rT + EG \cdot (rT - 1) \quad (2.3.7)$$

$$argtr = XTIR \cdot \ln rT \quad (2.3.8)$$

Temperature dependence of various current parameters is given by,

$$IS_t = IS \cdot \exp(argt) \quad (2.3.9)$$

$$ISR_t = ISR \cdot \exp(argtr) \quad (2.3.10)$$

$$ISE_t = ISE \cdot \exp(argt/NE) \quad (2.3.11)$$

$$ISC_t = ISC \cdot \exp(argt/NC) \quad (2.3.12)$$

Temperature dependence of parameters related to post reverse breakdown I-V model is given by following equations.

$$IJBV_t = IJBV \cdot (1 + XJBV \cdot (rT - 1)) \quad (2.3.13)$$

$$BVR_t = BVR \cdot (1 + XBVR \cdot (rT - 1)) \quad (2.3.14)$$

$$IJBVC_t = IJBVC \cdot (1 + XJBVC \cdot (rT - 1)) \quad (2.3.15)$$

Temperature dependence of resistances are modeled with equations below.

$$RB_t = RB \cdot rT^{ARB} \quad (2.3.16)$$

$$RE_t = RE \cdot rT^{ARE} \quad (2.3.17)$$

$$RC_t = RC \cdot rT^{ARC} \quad (2.3.18)$$

Temperature dependence of junction capacitance is modeled with the following equations. These temperature dependence formulations are applied to all junction capacitances in the model with corresponding change to MJ and VJ parameters.

$$F_1 = \frac{T_{nom}}{Ref_t} \quad (2.3.19)$$

$$F_2 = \frac{T_{dev}}{Ref_t} \quad (2.3.20)$$

$$E_{g,f} = E_{g0} - \frac{egta \cdot T_{dev}^2}{egtb + T_{dev}} \quad (2.3.21)$$

$$A_{rg0} = \frac{-E_{g,f}}{2 \cdot K_b \cdot T_{dev}} + \frac{E_{g,300}}{2 \cdot K_b \cdot Ref_t} \quad (2.3.22)$$

$$P_f = \frac{-2V_t}{(3/2)\ln(F_2) + q \cdot A_{rg0}} \quad (2.3.23)$$

$$P_0 = \frac{VJ - P_f}{F_1} \quad (2.3.24)$$

$$G_{m0} = \frac{VJ - P_0}{P_0} \quad (2.3.25)$$

$$C_{j,t} = \frac{CJ}{1 + MJ(T_{nom} - Ref_t) \cdot 4e - 4 - G_{m0}} \quad (2.3.26)$$

$$V_{j,t} = F_2 \cdot P_0 + P_f \quad (2.3.27)$$

$$G_{m,n} = \frac{V_{j,t} - P_0}{P_0} \quad (2.3.28)$$

$$C_{j,t,final} = \frac{C_{j,t}}{1 + MJ(T_{nom} - Ref_t) \cdot 4e - 4 - G_{m,n}} \quad (2.3.29)$$

The ambient temperature dependence on parameters is described by the above equations. However, for better readability of the equations the temperature dependent variable calculated by applying the above equations to core parameter is not used in the following sections. For example, BF is used instead of BF_t and likewise for other parameters.

2.4 Current calculations

The total emitter current (see Fig. 2) is sum of base, and collector current. Base current I_b is given by,

$$I_b = I_{be} + I_{bc} \quad (2.4.1)$$

where I_{be} and I_{bc} are the currents in base-emitter and base-collector junctions respectively. I_{be} is the active current for usual ESD diode operation and modeling. I_b is further formulated as,

$$I_{be} = \frac{I_f}{BF} + I_{be,r} \quad (2.4.2)$$

$$I_f = I_1 - I_2 \quad (2.4.3)$$

$$I_1 = IS \cdot \left(\exp\left(\frac{V_{be,i}}{NF \cdot V_t}\right) - 1 \right) \quad (2.4.4)$$

$$I_2 = \frac{IJBV}{1 + THER \cdot V_{be,i}^{THEEXP}} \cdot \left(\exp\left(\frac{-V_{be,i} - BVR}{NBV \cdot V_t}\right) - 1 \right) \quad (2.4.5)$$

I_f models the forward, reverse breakdown, and post-breakdown behavior of the ESD diode. $I_{BE,r}$ models tunneling and recombination current under the conditions of small reverse bias. It is formulated as,

$$I_{be,r} = ISR \cdot \left(\exp\left(\frac{-V_{be,i} \cdot VTR}{NTR \cdot V_t \cdot (VTR - V_{be,i})}\right) - 1 \right) + ISE \cdot \left(\exp\left(\frac{V_{be,i}}{NE \cdot V_t}\right) - 1 \right) \quad (2.4.6)$$

Equivalent to Eqs. 2.4.2 - 2.4.6 for I_f and $I_{BE,r}$ following formulations exist for base-collector junction current.

$$I_{bc} = \frac{I_r}{BR} + I_{bc,r} \quad (2.4.7)$$

$$I_r = I_3 - I_4 \quad (2.4.8)$$

$$I_3 = IS \cdot \left(\exp\left(\frac{V_{bc,i}}{NR \cdot V_t}\right) - 1 \right) \quad (2.4.9)$$

$$I_4 = \frac{IJBVC}{1 + THER \cdot V_{bc,i}^{THEEXP}} \cdot \left(\exp\left(\frac{-V_{bc,i} - BVR}{NBVC \cdot V_t}\right) - 1 \right) \quad (2.4.10)$$

$$I_{bc,r} = ISR \cdot \left(\exp\left(\frac{-V_{bc,i} \cdot VTRC}{NTRC \cdot V_t \cdot (VTRC - V_{bc,i})}\right) - 1 \right) + ISC \cdot \left(\exp\left(\frac{V_{bc,i}}{NC \cdot V_t}\right) - 1 \right) \quad (2.4.11)$$

The collector current is given by,

$$I_c = I_{tf} - I_{tr} \quad (2.4.12)$$

$$I_{tf} = I_f \cdot I_{k1} \quad (2.4.13)$$

$$I_{tr} = I_r \cdot I_{k1} \quad (2.4.14)$$

$$I_{k1} = 2 \cdot \frac{1 - \frac{V_{be,i}}{V_{AR}} - \frac{V_{bc,i}}{V_{AF}}}{1 + T_0^{XKF}} \quad (2.4.15)$$

$$T_0 = 1 + 4 \left(\frac{I_f}{IKF} \cdot (1 + V_{bici} \cdot IKBMW) + \frac{I_r}{IKR} \right) \quad (2.4.16)$$

I_{k1} accounts for the high level injection and BJT gain reduction.

The delayed response in collector is modeled with the following formulations. Following differential equation is solved in the model for calculating a delayed current ratio.

$$(V_{biei} - V_{tbb}) + 10^{-6} \cdot V_{tbb} + CTHBB \cdot \frac{V_{tbb}}{dt} = 0 \quad (2.4.17)$$

V_{tbb} is used to calculate the delayed current ratio.

$$d_{ratio} = \frac{\min(V_{biei}, V_{tbb})}{\max(V_{biei})} \quad (2.4.18)$$

I_{tf} in the collector current equation is modified to the following.

$$I_{tf} = I_{tf} \cdot d_{ratio} \cdot CDELAY + (1 - CDELAY) \cdot I_{tf} \quad (2.4.19)$$

2.5 Base resistance formulations

High field, and the effect of charge modulation is modeled with following formulations.

$$R_b = RB \cdot \frac{T_1^{1/MEXP}}{1 + T_2} \quad (2.5.1)$$

$$T_1 = 1 + (\frac{V_{bb,i}}{V_{SATB}})^{MEXP} \quad (2.5.2)$$

$$T_2 = (\frac{Q_m}{Q_{TT0}})^{QEXP} \quad (2.5.3)$$

Q_m is obtained by solving the differential equation Eq. 2.5.4 in the model.

$$\frac{dQ_m}{dt} + \frac{Q_m}{T_f} - \frac{I_f}{BF} = 0 \quad (2.5.4)$$

The effect of mobility degradation under the influence of high-field on transit time (T_f) is modeled with,

$$T_f = \frac{L_p^2}{D} \quad (2.5.5)$$

where L_p is the length, and D is the diffusion constant. Considering the relationship between D and mobility to be,

$$D = \mu_{eff} \frac{kT}{q} \quad (2.5.6)$$

where μ_{eff} includes the effect of electric field on the mobility with

$$\mu_{eff} = \frac{\mu_0}{1 + (\mu_0 \cdot E / v_{sat})^{TEXP}} \quad (2.5.7)$$

To simplify the implementation, the field dependence of transit time due to mobility variation is modeled by combining equations Eq. 2.5.5 to Eq. 2.5.7 resulting in,

$$V_{tff} = (\frac{V_{be}}{V_{TF0}})^{TEXP} \quad (2.5.8)$$

$$V_{tff1} = (1 + V_{tff})^{1/TEXP} - 1 \quad (2.5.9)$$

$$T_f = TF \cdot (1 + ATFF \cdot V_{tff1}) \quad (2.5.10)$$

2.6 Capacitance formulations

The capacitance of the junctions have been implemented with formulation for the junction charges. The junction charge formulations lead to following equations for the space charge and diffusion capacitors.

$$C_{be} = \frac{CJE}{(1 - (\frac{V_{be,i}}{V_{JE}})^{MJE})} + T_f \cdot \frac{dI_f}{dV_{bei}} \quad (2.6.1)$$

$$C_{bc} = \frac{CJC}{(1 - (\frac{V_{bc,i}}{V_{JC}})^{MJC})} + T_r \cdot \frac{dI_r}{dV_{bci}} \quad (2.6.2)$$

2.7 Self Heating Model

The self-heating effect is modeled using an R-C network modeling the thermal impedance of the device. There are two different thermal networks implemented in the model which can be chosen based on the model switch SHMOD. The thermal network for SHMOD = 1 is shown in Fig. 4.

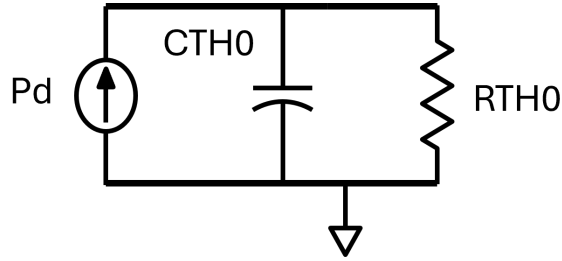


Figure 4: R-C thermal network for self-heating effect when SHMOD=1.

The thermal network in the model for SHMOD=2 is shown in Fig. 5.

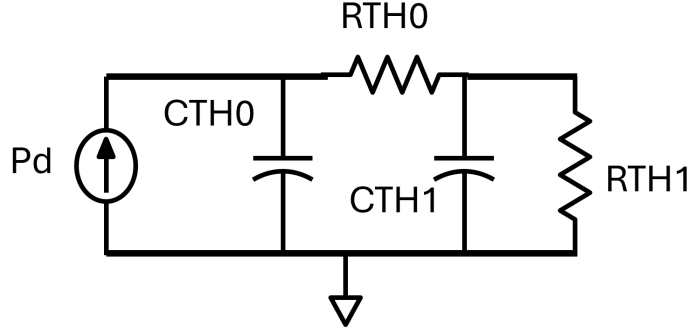


Figure 5: R-C thermal network for self-heating effect when SHMOD=2.

The voltage at DT node in Fig. 4 and Fig. 5 represent the local device temperature rise in the device and is used in model calculations.

2.8 Failure warning

Failure temperature can be set in the ASM-ESD model with model parameter TFAIL. During simulations when the highest temperature in the device modelled via the thermal network reaches TFAIL a failure warning is reported by the model. The default values of TFAIL is set to 1000 K. Thermal and Flicker noise models are included in ASM-HEMT model.

2.9 Flicker Noise Model

Our unified flicker noise model takes both the theories of mobility fluctuation and carrier number fluctuation into account. The power spectral density (PSD) of the mean square fluctuations in number of occupied traps is given as,

$$S_N(f) = KF \cdot \frac{I_{be}^{AF}}{f} \quad (2.9.1)$$

2.10 Thermal Noise Model

The thermal noise of all three resistances is included in the model with the following equations. The thermal noise is white in nature. For base resistance R_b ,

$$S_{R_b} = \frac{4 \cdot k \cdot}{R_b} \quad (2.10.1)$$

$$S_{R_c} = \frac{4 \cdot k \cdot}{R_c} \quad (2.10.2)$$

$$S_{R_e} = \frac{4 \cdot k \cdot}{R_e} \quad (2.10.3)$$

2.11 Note on device size scaling in the model

The formulations in ASM-ESD model scale with finger-length (L) and number of fingers (N). However, the scaling in ASM-ESD assumes linear and uniform change in electrical behavior with the device size. Any layout dependent effects, non-uniform current distributions across fingers, scaling of thermal effects due to layout are not part of the formulations due to their technology-specific nature. These effects can be included by creating a wrapper or sub-circuit around the ASM-ESD model, for instance layout parasitic extracted with the help of parasitic extraction tools. The layout dependent scaling of ASM-ESD model parameters can then be included in the wrapper.

3 Parameter List

3.1 Model Controllers

Name	Unit	Default	Min	Max	Description
RBMOD	-	1	0	1	Switch to select base resistance model. 0: Fixed 1: Bias dependent
SHMOD	-	2	0	2	Switch to turn OFF/ON the self-heating effect model;0 to turn off,1 to turn on one time-constant (RC) network, 2 to turn on two time-constants models.
EXTMOD	-	0	0	1	Switch to turn-on/off extrinsic p-cell parasitic model; 0 to not include P-cell parasitic elements and 1 to include.
TYPE	-	1	1	1	Device type 1:NPN -1:PNP.

3.2 Device size parameters

Name	Unit	Default	Min	Max	Description
L	m	10e-6	20e-9	∞	Finger length
N	-	1	1	∞	Number of fingers.

3.3 Current Model Parameters

Name	Unit	Default	Min	Max	Description
IS	A/m	1e-17	0	1	Saturation current
ISC	A/m	0	0	1	B-C leakage Saturation current
BF	-	0.1	> 0	∞	Forward BJT current gain
BR	-	0.1	> 0	∞	Reverse BJT current gain
NF	-	1.0	> 0	∞	Forward current emission coefficient

NR	-	1.0	> 0	∞	Forward emission coefficient for B-C junction
NC	-	2.0	0	∞	Emission coefficient for B-C current.
VAF	V	0.0	0	∞	Forward early voltage parameter
IKF	A/m	0	0	1000	Forward gain high current roll-off parameter.
XKF	-	0.9	0	∞	Forward BJT gain roll-off at high current exponent parameter.
IKR	A/m	0	0	1000	Reverse gain high current roll-off parameter.
ISR	A/m	0	0	1	Tunnel current coefficient.
ISE	A/m	0	0	1	B-E leakage saturation current.
NE	-	1.5	> 0	10	B-E leakage emission coefficient.
NTR	-	5	0	500	Reverse tunnel emission coefficient
VTR	V	10	> 0	∞	Reverse tunnel current bias dependence parameter.
VAR	V	0	0	∞	Reverse early voltage parameter.
BVR	V	10	1	∞	Reverse breakdown voltage.
IJBV	A/m	5	0	∞	Post reverse breakdown current parameter for B-E junction.
IJBVC	A/m	5	0	∞	Post reverse breakdown current multiplier for B-C junction.
NBV	-	10	> 0	500	Slope of current near breakdown for B-E junction.
NBVC	-	10	> 0	500	Slope of current near breakdown for B-C junction.
KBWM	$1/V^{XBWM}$	0	0	500	Base-width modulation impact on BJT gain
XBWM	-	1	0	500	Exponent for tuning effect of base-width modulation on BJT gain
IKBWM	$1/V$	0	0	∞	Base width modulation multiplier parameter

CTHBB	-	1e-9	> 0	∞	Delayed BJT collector current time-constant parameter
CDELAY	-	0	0	1	Delayed BJT current parameter, set 0 for no delay.
THER	$1/V^{THEEXP}$	0.01	0	∞	Post reverse breakdown I-V model parameter.
THEEXP	-	1.1	0	∞	Post reverse breakdown resistance non-linearity parameter.
EG	eV	1.1	> 0	∞	Band gap.

3.4 Resistance Model Parameters

Name	Unit	Default	Min	Max	Description
RB	Ohm.m	1e-5	0	∞	Base resistance.
RC	Ohm.m	1e-6	0	∞	Collector resistance
RE	Ohm.m	1e-6	0	∞	Emitter resistance
RBE	Ohm.m	0	0	∞	Extrinsic base resistance, active when EXTMOD=1
RCE	Ohm.m	1e-6	0	∞	Extrinsic collector resistance, active when EXTMOD=1
REE	Ohm.m	1e-6	0	∞	Extrinsic emitter resistance, active when EXTMOD=1
TF	sec	0	0	∞	Forward transit time
PTF	-	0	0	∞	Excess phase at frequency $\frac{1}{\pi \cdot TF}$
TR	sec	0	0	∞	Reverse transit time
VTF0	V	100	≥ 0	∞	Carrier transit time field dependence threshold.
TEXP	V	2	0	∞	Carrier transit time field dependence parameter, active when RBMOD=1
ATFF	-	0	0	∞	Carrier transit time field dependence model multiplier.
MEXP	-	2	0	∞	Velocity saturation exponent parameter for base resistor.

MEXPE	-	2	0	∞	Velocity saturation exponent parameter for emitter resistance
QEXP	-	1	0	∞	Charge dependence of base resistance exponent parameter.
VSATB	V	2	0	∞	Voltage at which saturation starts in base resistor.
VSATE	V	2	0	∞	Voltage at which saturation starts in base resistor.
QTT0	C	20	> 0	∞	B-E junction capacitance exponential factor
FC	-	0.5	-20	20	Coefficient for forward bias depletion capacitance formula.
MINR	Ohm	1e-3	-	-	Minimum resistance parameter.

For parameter QTT0 an alias parameter VTT0 is also defined for backward compatibility of the model code.

3.5 Capacitance Parameters

Name	Unit	Default	Min	Max	Description
CJE	F/m	0	0	∞	Zero bias B-E junction capacitance
MJE	-	0.33	0	∞	B-E junction capacitance exponential factor.
VJE	V	0.75	0	∞	B-E junction built-in potential.
CJC	F/m	0	0	∞	Zero bias B-C junction capacitance
MJC	-	0.33	0	∞	B-C junction exponential factor.
VJC	V	0.75	0	∞	B-C junction built-in potential.
CJS	F/m	0	0	∞	Zero bias E-C junction capacitance
MJS	-	0.33	0	∞	E-C junction exponential factor.
VJS	V	0.75	0	∞	E-C junction built-in potential.
XCJC	-	0.9	0	∞	Fraction of depletion B-C capacitance connected to internal B-C node.

3.6 Noise Model Parameters

Name	Unit	Default	Min	Max	Description
KF	-	0	0	∞	Flicker Noise coefficient.
AF	-	1	0	∞	Flicker noise exponent parameter.

3.7 Temperature Dependent and Self-Heating Parameters

Name	Unit	Default	Min	Max	Description
XTB	-	0	-10	10	Temperature Dependence of BF and BR.
TNOM	deg C	25	-40	125	Nominal temperature for parameter extraction.
XTI	-	3	0	20	Temperature dependence of IS.
XTIR	-	3	0	20	Temperature dependence of ISR.
RTH0	<i>Ohm</i>	5e-4	0	∞	Thermal Resistance active in SHMOD=1 and 2
CTH0	<i>F</i>	1e-4	0	∞	Thermal Resistance active in SHMOD=1 and 2
RTH1	<i>Ohm</i>	5e-7	0	∞	Thermal Resistance active in SHMOD=2
CTH1	<i>F</i>	1e-7	0	∞	Thermal Resistance active in SHMOD=2
TFAIL	K	1000	0	∞	Failure temperature.
XJBV	-	0	0	∞	Temperature dependence of IJBV.
XJBVC	-	0	0	∞	Temperature dependence of IJBVC.
XBVR	-	0	0	∞	Temperature dependence of BVR.
XTHEEXP	-	0	$-\infty$	∞	Temperature dependence of THEEXP.
ARB	-	0	0	∞	Temperature dependence of RB.
ARC	-	0	0	∞	Temperature dependence of RC.
ARE	-	0	0	∞	Temperature dependence of RE.
DTEMP	deg C	0	$-\infty$	∞	Temperature variability parameter.

4 Parameter Extraction

4.1 I-V model parameters

I-V model results achieved during phase II are shown in Fig. 6. The model parameters exercised for these I-V model results are:

- Forward I-V parameters: I_S , N_F , R_B , R_E , R_C . I_S is extracted from the magnitude of the forward current and N_F can be extracted from the slope of logarithmic of current with applied voltage. I_S and N_F are extracted from regions where diode current shows exponential dependence on the applied voltage. At higher voltages, the resistance parameters R_B , R_E , and R_C affect the current.
- Reverse I-V parameters: I_{SATR} , V_{TR} , N_{TR} . I_{SATR} can be extracted from the magnitude of the reverse current. Increasing(decreasing) I_{SATR} will proportionally increase(decrease) the reverse bias current. V_{TR} can be extracted from region where reverse bias current shows an increase with the applied reverse voltage. N_{TR} can be extracted from the slope of reverse current increase.
- Temperature dependence parameters: X_{TI} , X_{TIR} . X_{TI} can be extracted from the temperature dependence of forward current while X_{TIR} can be extracted from the temperature dependence of the reverse bias diode current.

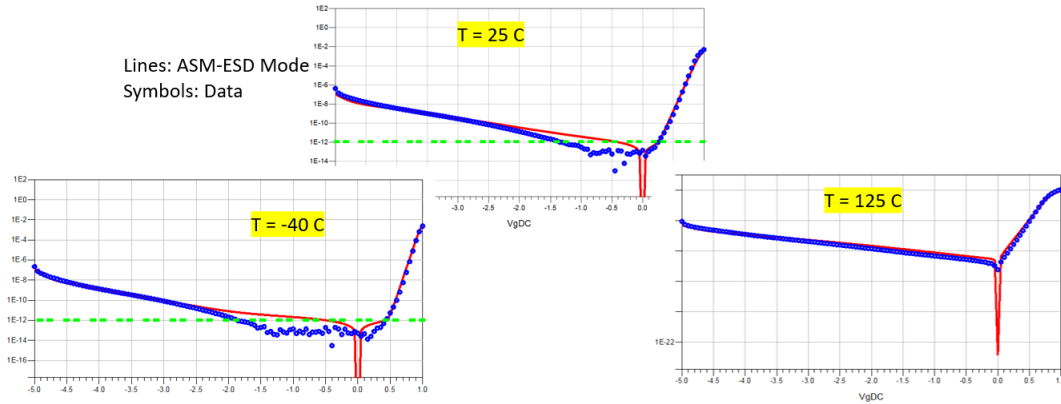


Figure 6: Forward and reverse I-V model results.

4.2 C-V model parameters

The model parameters exercised for C-V model shown in Fig. 7 are: CJE, VJE, and MJE. CJE controls the level of junction capacitance. VJE controls the bias dependence, setting VJE to a very large value will eliminate the bias dependence. MJE controls the slope/shape capacitance versus applied voltage curve.

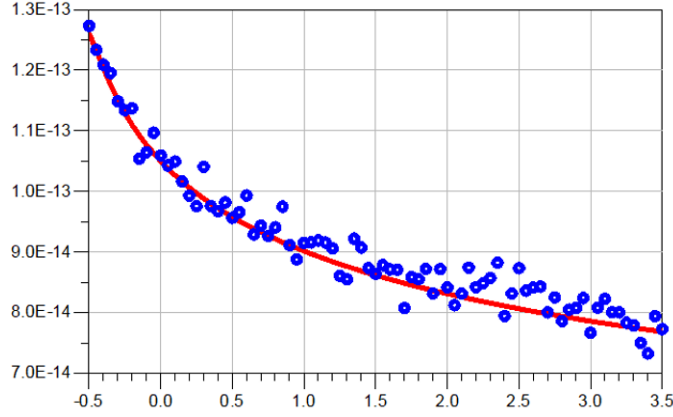


Figure 7: C-V model results.

4.3 Post breakdown resistance model parameters

The model parameter exercised for modeling the resistance of the diode after the reverse breakdown are: BV, NBV, IJBV, THER, THEREXP. IJBV models the magnitude of the post-reverse breakdown current. BV models the breakdown voltage. NBV controls the slope of current rise near and just after breakdown. THER and THEREXP control the non-linearity seen in the post-reverse breakdown current with applied voltage. Setting THER to zero will remove this non-linearity. THER and THEREXP provide flexibility in modeling post-reverse breakdown non-linearity in current. Temperature dependence of post reverse breakdown model XJBV, XTHEREXP, and XBVR have been used to model temperature dependence of this behavior.

4.4 Overshoot model parameters

To model the overshoot effects following parameters can be exercised:

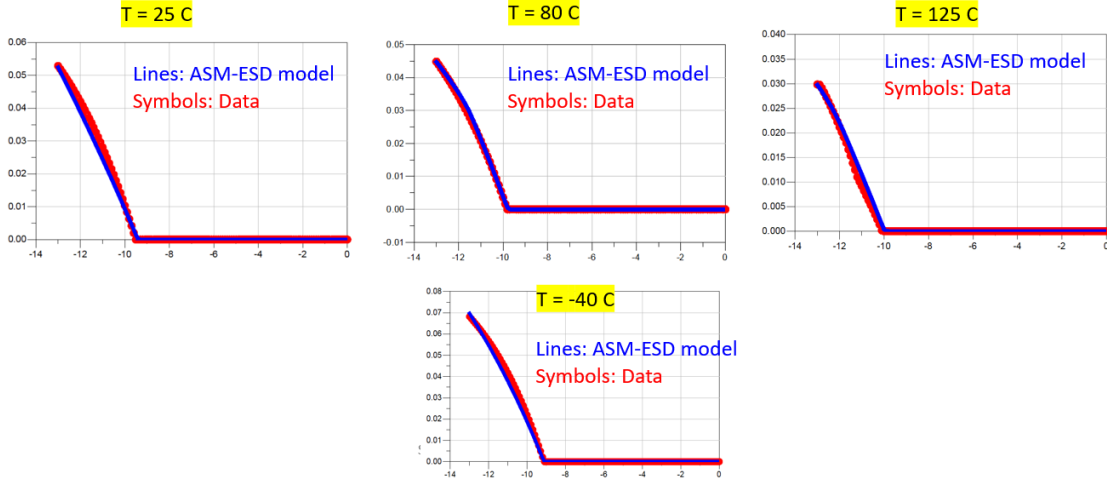


Figure 8: Post reverse breakdown resistance model results.

- Transit time parameters: T_F , $ATFF$, $VTF0$, and $TEXP$. T_F models the transit time. It can be extracted from the observed overshoot duration in the transient measurements. $VTF0$ and $ATFF$ can be used to model the variation in transit time with pulse magnitude. $VTF0$ controls the bias condition above which transit time shows bias dependence. $ATFF$ and $TEXP$ can be used to model the bias dependence of transit time.
- Base resistance parameters: RB , $VTT0$, $VSATB$. RB models the base resistance at the beginning of the pulse. $VTT0$ can be used to model the decrease in base resistance as charge flows into the region. Decreasing $VTT0$ will reduce the base resistance towards the end of the pulse. $QEXP$ models non-linearity in reduction of base resistance towards end of the pulse and is useful when modeling base resistance change with multiple pulse amplitudes.

Beyond the overshoot the self-heating effect seen in the transient data is modeled with the thermal model parameters $RTH0$, $CTH0$, $RTH1$, and $CTH1$. The parameters modeling the temperature dependence of base-, emitter-, and collector resistances are also used. These are ARB , ARE , and ARC respectively.

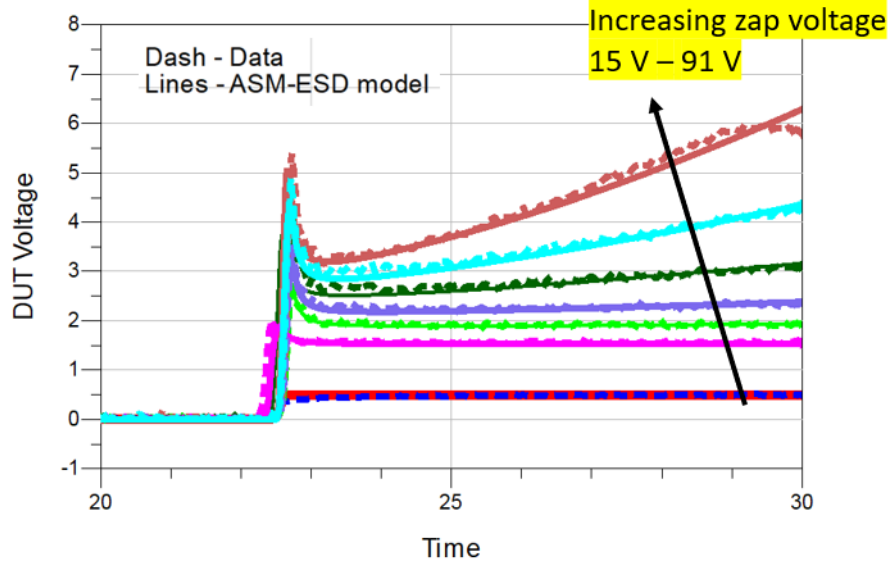


Figure 9: Overshoot model results.

4.5 BJT Gain model parameters

The gain of the parasitic BJT is modeled with parameter BF. The roll-off of gain with current is modeled with parameters IKF and XKF. BF parameter can be obtained from modeling the collector current. Plotting IB and IC in form of Gummel plot is very useful to extract current parameters and BF. IKF models the forward current level where high-level injection results in an increased roll-off of current and as a result a reduction in gain BF with current. XKF is used to model the slope of the roll-off in BF.

4.6 BJT Gain V_{cb} dependence

Change in BJT gain with V_{cb} is modeled with parameters VAF, KBWM, and XBWM. KBWM is multiplier parameter which turns on/off the V_{cb} dependence of BJT gain. XBWM can be used to control the nature of BJT gain with V_{cb} , setting XBWM to unity will result in linear dependence.

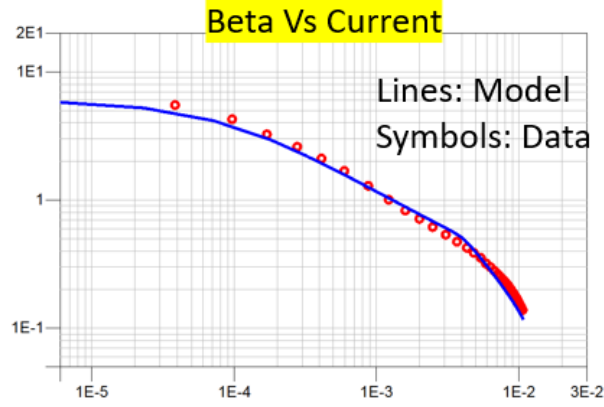


Figure 10: Parasitic BJT model results.

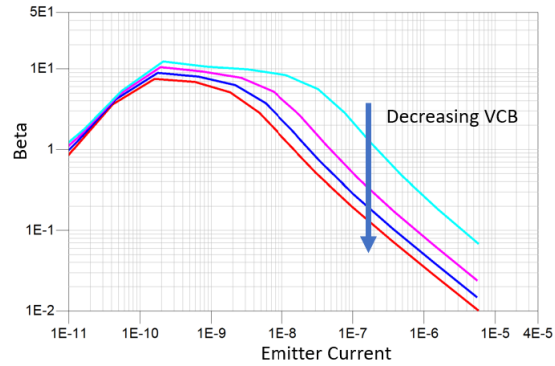


Figure 11: BJT gain V_{cb} dependence.

4.7 Delayed BJT response model parameters

The delayed response of BJT is modeled show in Fig. 12 with a separate time-constant. This is modeled with parameter CTHBB and the model parameter ABETA. CTHBB controls the time-constant of transient behavior of BJT gain. ABETA is a multiplier parameter which can turn on/off the transient effect in BJT gain.

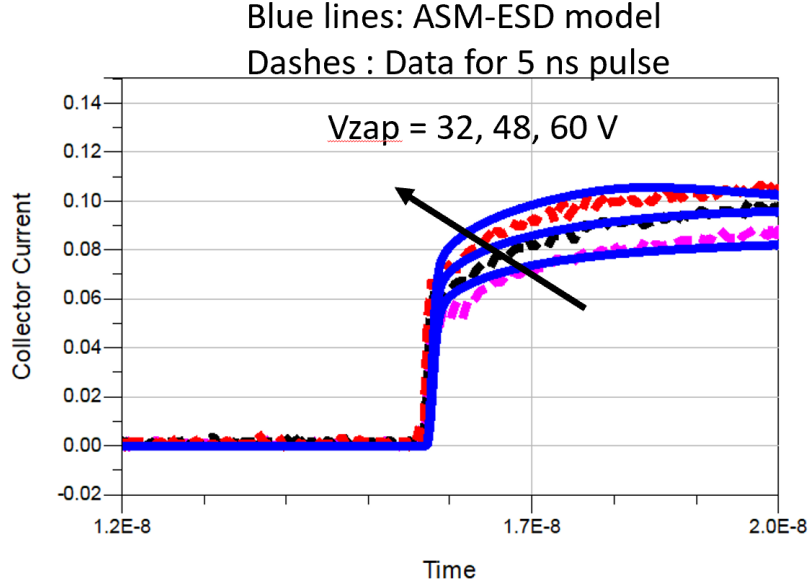


Figure 12: Delayed BJT response model results.

5 Acknowledgements

We would like to thank all the CMC member companies for their interest in this model. We acknowledge very fruitful technical discussions with ESD work-group members, especially Michael Stockinger (NXP), Paul Zhou (Analog Devices), Meng Miao (Globalfoundries), Geoffrey Coram (Analog Devices), and You Li (with Globalfoundries at the time).

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Manual created: April 5, 2025